

Bluetooth Series

Data Sheet

USE_8761

Bluetooth Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	07/09/2026	MB	LB	Initial release

Contents

Features :	3
General.....	3
MCU Platform.....	3
Host Interface.....	3
Bluetooth Features.....	3
General Description:	4
Pinout Diagram:	4
Pin Descriptions:	5
Electrical Specifications :	6
Power Supply DC Characteristics:.....	6
Power Consumption :	6
RF Characteristics.....	6
Bluetooth Specification.....	6
Block Diagram :	8
Reference Design :	8
Recommended Footprint Dimensions:	9
Module Dimension Drawing:	10
Product Dimension:	10
Recommended Reflow Profile :	10

Features :

General

- Supports Bluetooth 5.0 functionalities
- Supports all packet types in Basic Rate and Enhanced Data Rate
- Supports Secure Simple Pairing
- Supports multiple Bluetooth Low Energy states
- Dimension: 13 × 15 mm
- Operating temperature: 0°C to 70°C
- Storage temperature: -40°C to 85°C

MCU Platform

- Chipset: RTL8761BTV

Host Interface

- UART interface with configurable baud rate
- PCM interface

Bluetooth Features

- Bluetooth 5.0 compliant
- Supports Bluetooth Classic (BDR/EDR)
- Supports Bluetooth Low Energy (BLE)
- Supports simultaneous BR/EDR and Low Energy links
- Compatible with Bluetooth v2.1 + LE systems
- Supports multiple active BR/EDR links in master or slave mode
- Supports multiple active Low Energy links in slave mode

General Description:

The USE_8761 is a low-cost and low-power Bluetooth module based on the highly integrated RTL8761BTV single-chip Bluetooth 5.0 controller. It provides UART and PCM interfaces and complies with the Bluetooth Core Specification version 5.0.

The module supports Bluetooth Dual Mode operation, including Bluetooth Classic BR/EDR and Bluetooth Low Energy. It is also compatible with previous Bluetooth versions, including Bluetooth v2.1 + LE.

For BR/EDR operation, the module supports multiple active links in either master or slave mode. For Bluetooth Low Energy operation, it supports multiple states and multiple active links in slave mode. BR/EDR and Low Energy links can remain active simultaneously.

The USE_8761 is designed for portable devices and Bluetooth controller applications.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

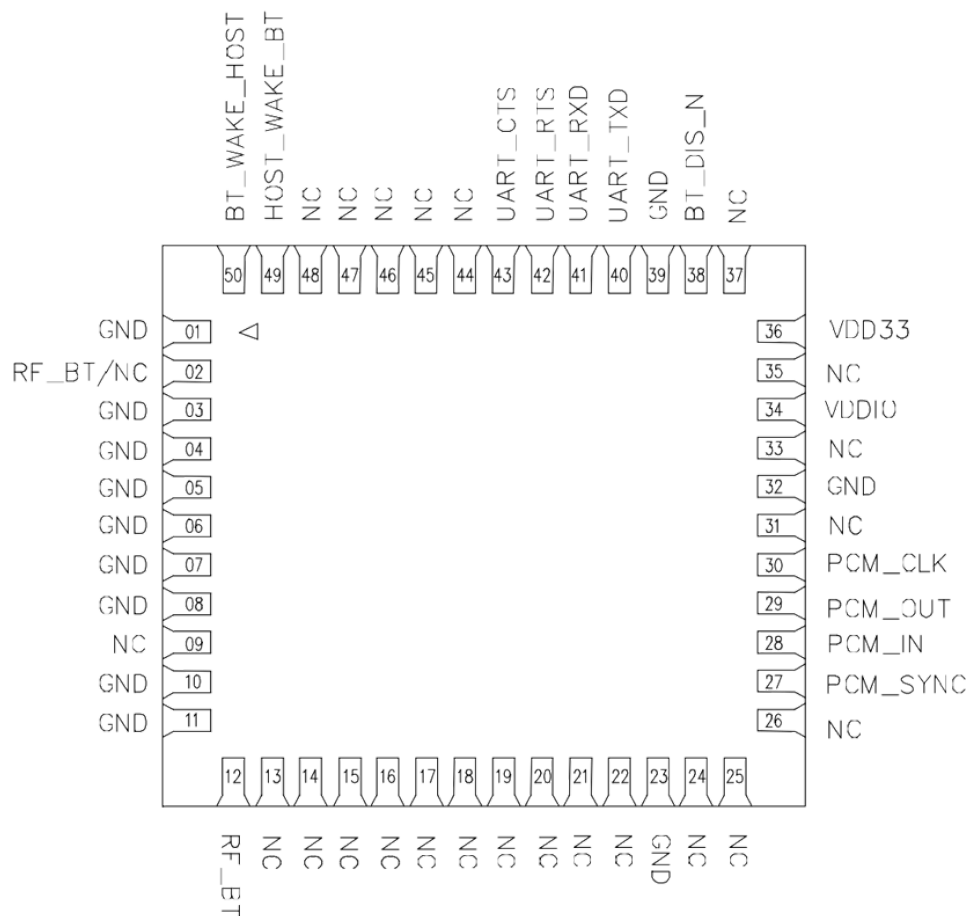


Fig1: Module Pinouts

Pin Descriptions:

Pin No	Pin Name	Type	Description	Voltage
1	GND	—	Ground connections	
2	RF_BT/NC	—	BT RF IO or NC, default NC	
3~8	GND	—	Ground connections	
9	NC	—	No connection	
10,11	GND	—	Ground connections	
12	RF_BT		BT RF IO	
13~22	NC	—	No connection	
23	GND	—	Ground connections	
24~26	NC	—	No connection	
27	PCM_SYNC	I/O	PCM sync signal	VDDIO
28	PCM_IN	I	PCM data input	VDDIO
29	PCM_OUT	O	PCM Data output	VDDIO
30	PCM_CLK	I/O	PCM clock	VDDIO
31	NC	I	No connection	
32	GND	—	Ground connections	
33	NC	—	No connection	
34	VDDIO	P	VDDIO power input	3.3V/1.8V
35	NC	—	No connection	
36	VDD33	P	Main power source input	3.3V
37	NC	—	No connection	
38	BT_DIS_N	I	Enable pin for Bluetooth device ON: pull high; OFF: pull low Default high	3.3V
39	GND	—	Ground connections	
40	UART_TXD	O	Bluetooth UART interface	VDDIO
41	UART_RXD	I	Bluetooth UART interface	VDDIO
42	UART_RTS	O	Bluetooth UART interface	VDDIO
43	UART_CTS	I	Bluetooth UART interface	VDDIO
44~48	NC	—	No connection	
49	HOST_WAKE_BT	I	HOST wake-up Bluetooth device	VDDIO
50	BT_WAKE_HOST	O	Bluetooth device to wake-up HOST	VDDIO

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	0	25	70	°C
VDD33	3.0	3.3	3.6	V
VDDIO	1.8	3.3	3.63	V

Power Consumption :

Condition: VDD33 = 3.3V, VDDIO = 3.3V

Item	Mode	Current Value (mA)
1	DH5 TX @ 4dBm	53.1
2	2DH5 TX @ 4dBm	53.7
3	3DH5 TX @ 4dBm	53.7
4	BLE 1M TX @ 4dBm	46.71
5	DH5 RX	29.7
6	2DH5 RX	29.7
7	3DH5 RX	29.7
8	BLE RX	27.1

RF Characteristics

Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	Bluetooth V5.0 of 1, 2 and 3 Mbps.
Host Interface	UART
Antenna Reference	Small antennas with 0~2 dBi peak gain

Frequency Band	2402 MHz ~ 2480 MHz		
Number of Channels	79 channels for BDR+EDR 40 channels for BLE		
Modulation	GFSK, $\pi/4$ -DQPSK, 8-DPSK		
RF Specification			
	Min(dBm)	Typical(dBm)	Max(dBm)
BDR Output Power		4	
BLE Output Power		4	
Sensitivity @ BER=0.1% for GFSK (1Mbps)		-90	
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)		-85	
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)		-80	
Sensitive @PER=30.8% FOR BLE (1Mbps)		-90	
Maximum Input Level	GFSK (1Mbps):-20dBm		
	$\pi/4$ -DQPSK (2Mbps) :-20dBm		
	8DPSK (3Mbps) :-20dBm		

Block Diagram :

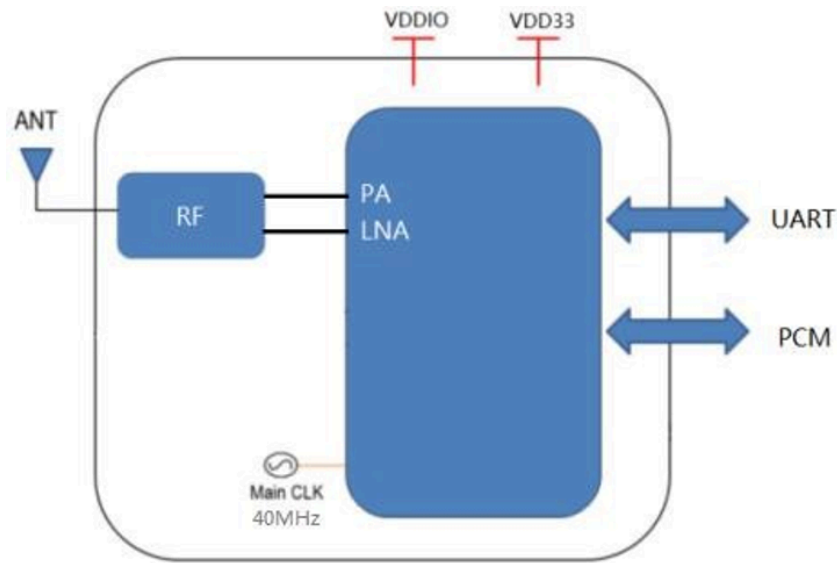


Fig 2 : Block Diagram

Reference Design :

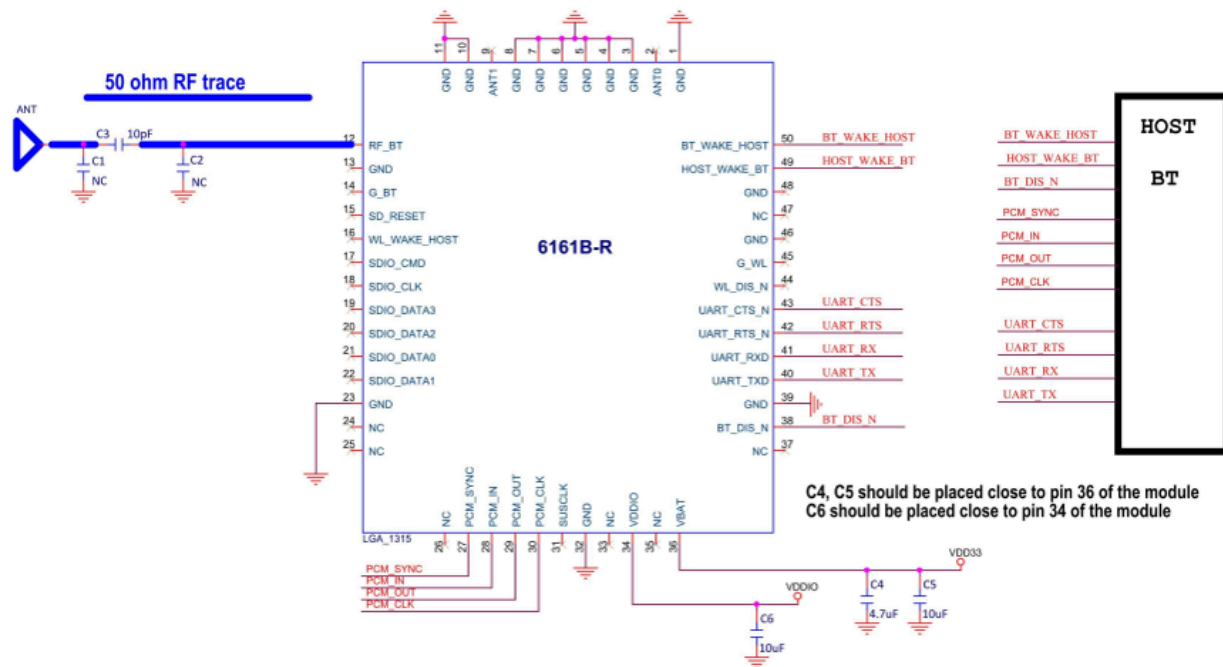


Fig 3 : Reference Design

Note: Module requires independent power supply , supply capacity $\geq 250\text{mA}$ and ripple less than 100mV; Do not share power with amplifier, infrared device, camera, etc.

Recommended Footprint Dimensions:

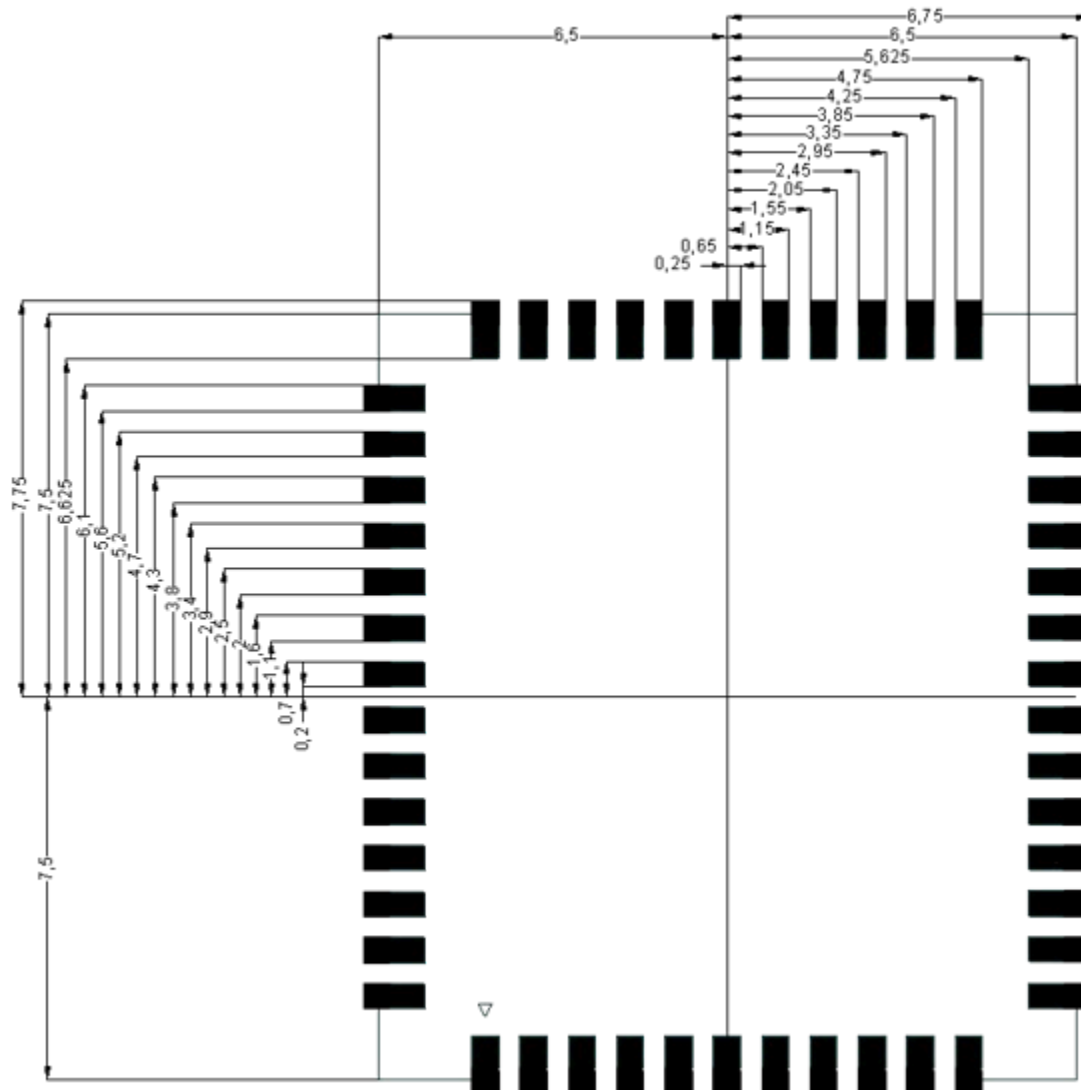


Fig 4 : Recommended Footprint Dimensions

Module Dimension Drawing:

- Module Dimension: L x W : 13 x 15 (+0.3/-0.1) mm

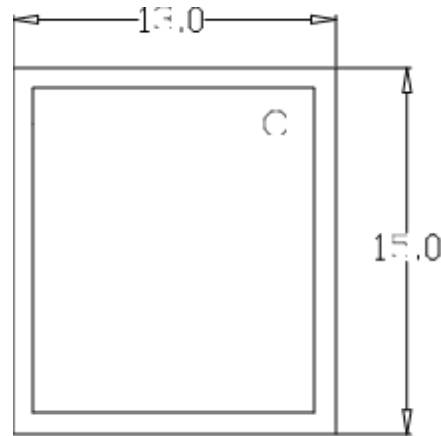


Fig 5(a) Module Dimension L × W

Product Dimension:

- H: 2. 25(±0.2)mm (Shielding version)
1. 65(±0.2)mm (No Shielding version)

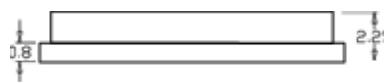


Fig 5(b) :Product Dimension

PCB thickness : 0.8 mm

Product Height :2. 25(±0.2)mm (Shielding version)

1. 65(±0.2)mm (No Shielding version)

Weight : 0.45±0.1g

Recommended Reflow Profile :

Referred to IPC/JEDEC standard. Peak

Temperature: 260±5 °C

5Time within 5°C of peak temperature : ≥10s

Number of Times : 2 times

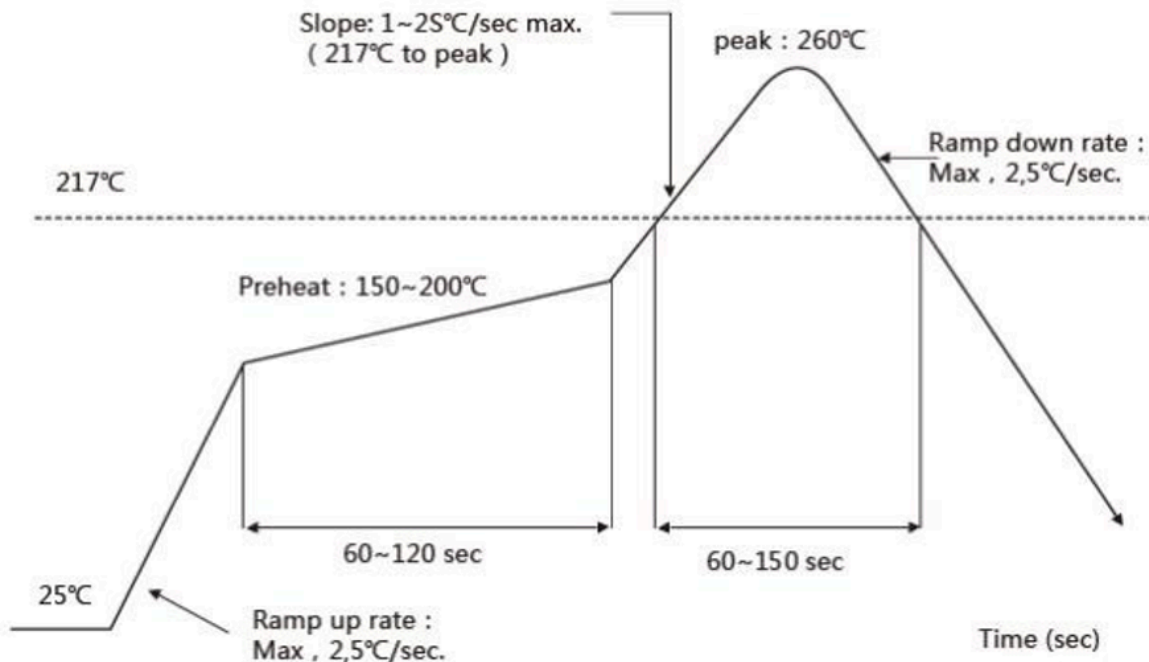
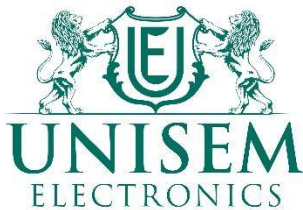


Fig 6 : Recommended Reflow Profile

Note:

1. The stay time in reflow zone(Temperature > 217°C) should no more than 120 seconds;
2. The peak temperature should no more than 260°C±5 °C;
3. The stay time in cooling zone should be greater than 10 seconds

For ordering USE_8761 please contact us on info@unisemindia.com.



Disclaimer and Copyright Notice

Unisem India Pvt Ltd or third parties may hold intellectual property rights in the products, names, logos and designs included in this document. Copying, reproduction, or modification of this document or any part thereof is only permitted with the express written permission of Unisem India Pvt Ltd. Disclosure to third parties is permitted for clearly public documents only. The information contained herein is provided "as is". No warranty of any kind, either express or implied, is made in relation to the accuracy, reliability, fitness for a particular purpose or content of this document. This document may be revised by Unisem India Pvt Ltd at any time. For most recent documents, please visit www.unisemholding.com.